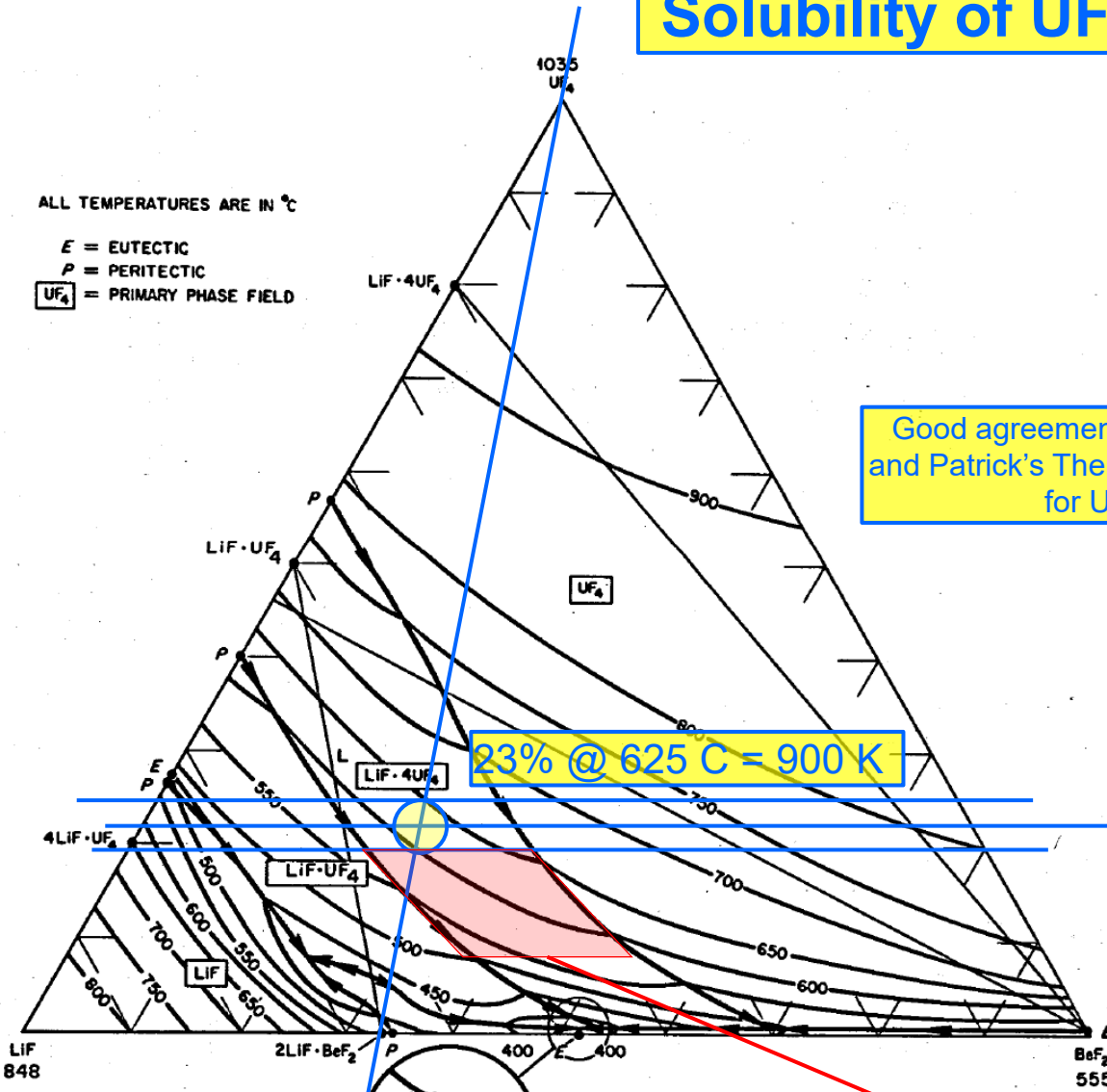


Solubility of UF₄



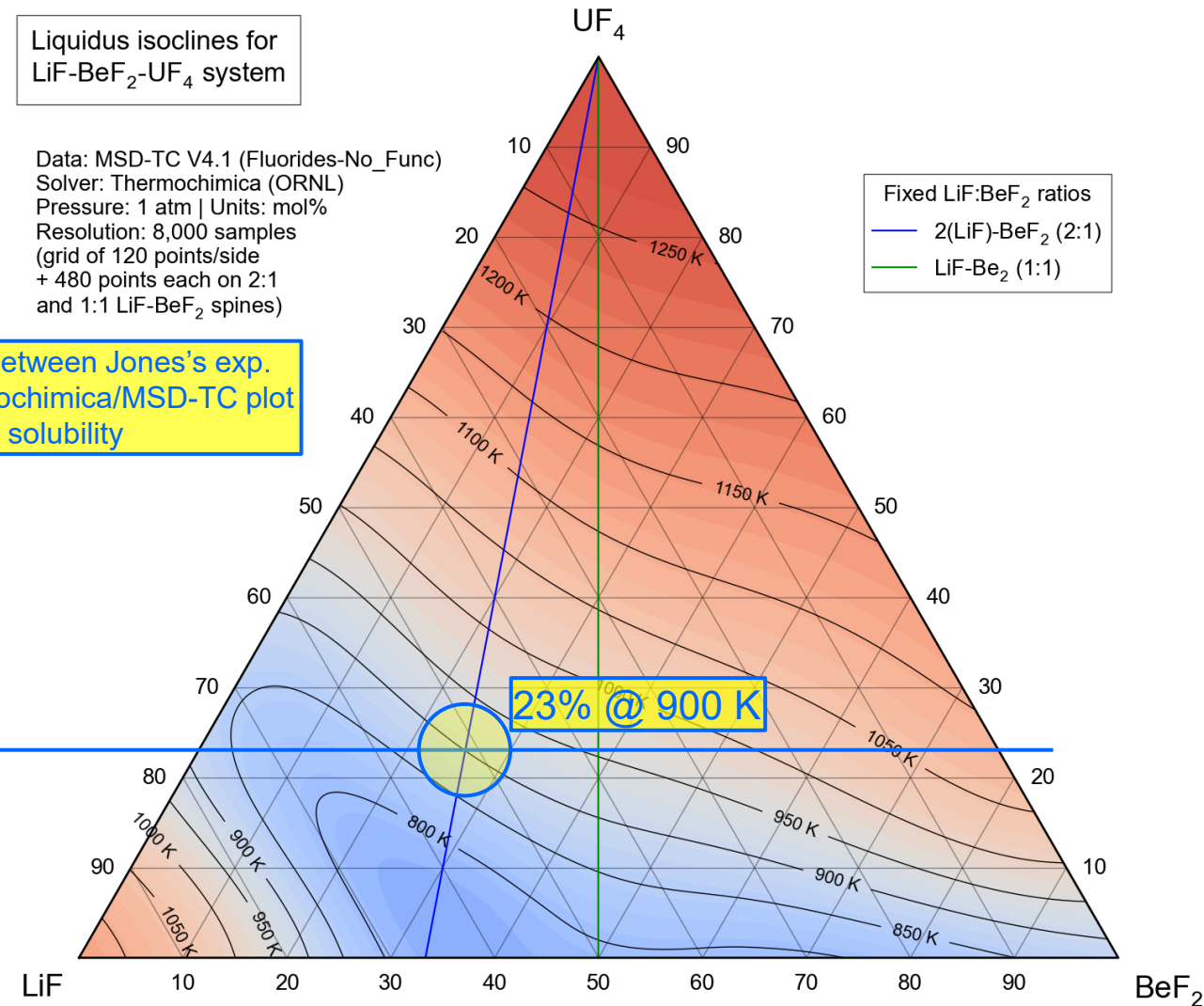
L. V. Jones's LiF-BeF₂-UF₄ ternary phase diagram

But Jones points out data in+near this band here were especially problematic because the LiF-4(UF₄) crystals are so heavy, they began physically segregating and smearing his probe's measurements. Thus, a different measurement method had to be used.

Liquidus isoclines for LiF-BeF₂-UF₄ system

Data: MSD-TC V4.1 (Fluorides-No_Func)
Solver: Thermochemica (ORNL)
Pressure: 1 atm | Units: mol%
Resolution: 8,000 samples
(grid of 120 points/side
+ 480 points each on 2:1
and 1:1 LiF-BeF₂ spines)

Good agreement between Jones's exp.
and Patrick's Thermochemica/MSD-TC plot
for UF₄ solubility



Patrick's diagram computed on 2026-04-12

van der Meer, J. Nucl. Mat. (2006):

- Used FactSage 5.3 solver with thermodynamic data from JRC-ITU (Karlsruhe/EU)
- Gets 30% UF₄ solubility in 2(LiF)-BeF₂ at 900 K
- vdMeer claims a difference of 20-70 K possible in measured T_{liq} between the two methods used by Jones 62 (see pg.53 of vdMeer's paper)
- Unclear what ORNL's MSD-TC considerations of this bias may be when fitting their models to Jones's data

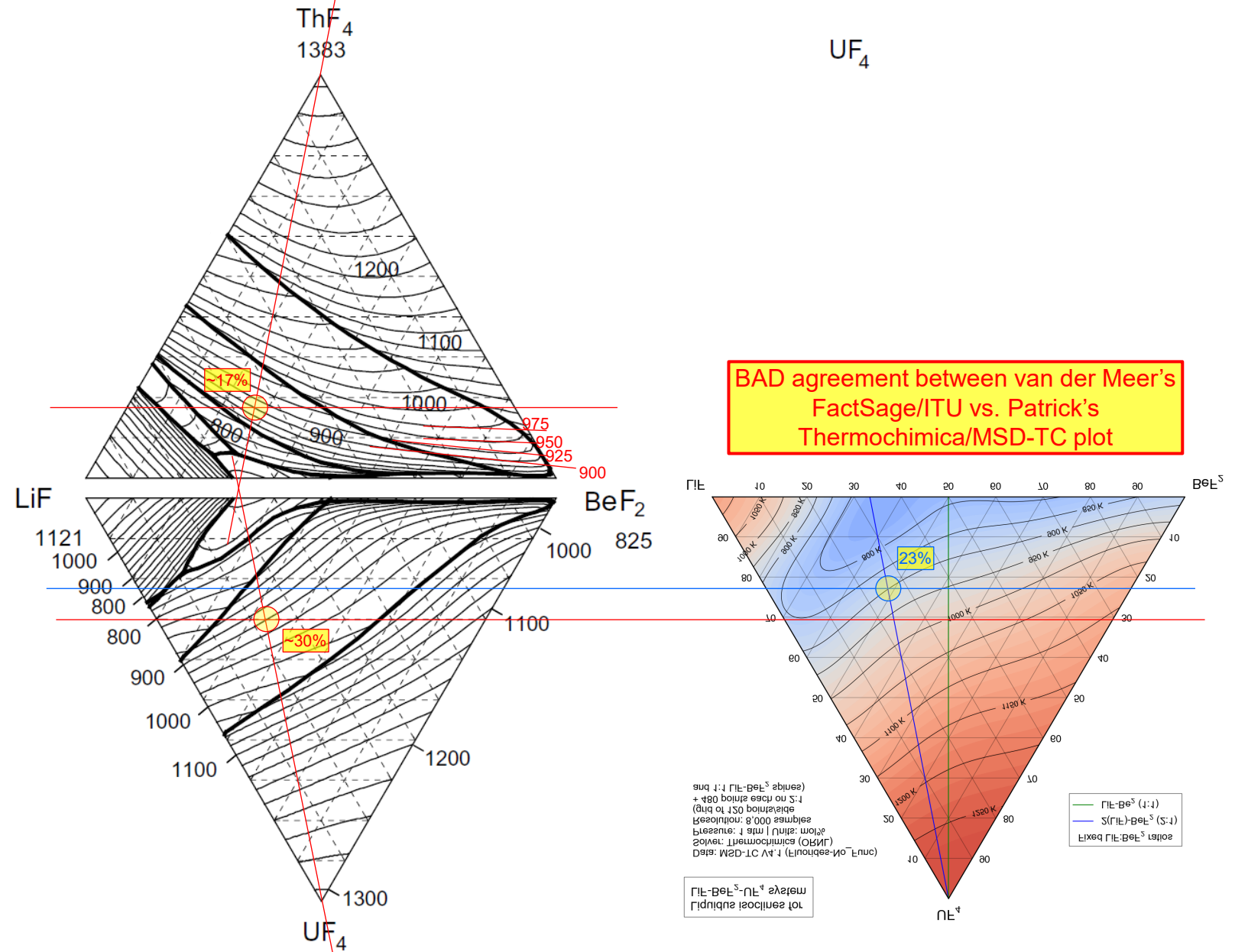
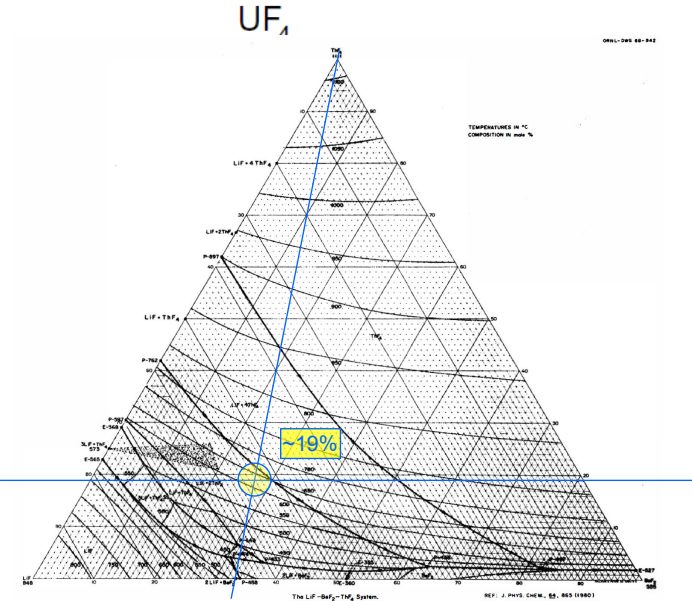
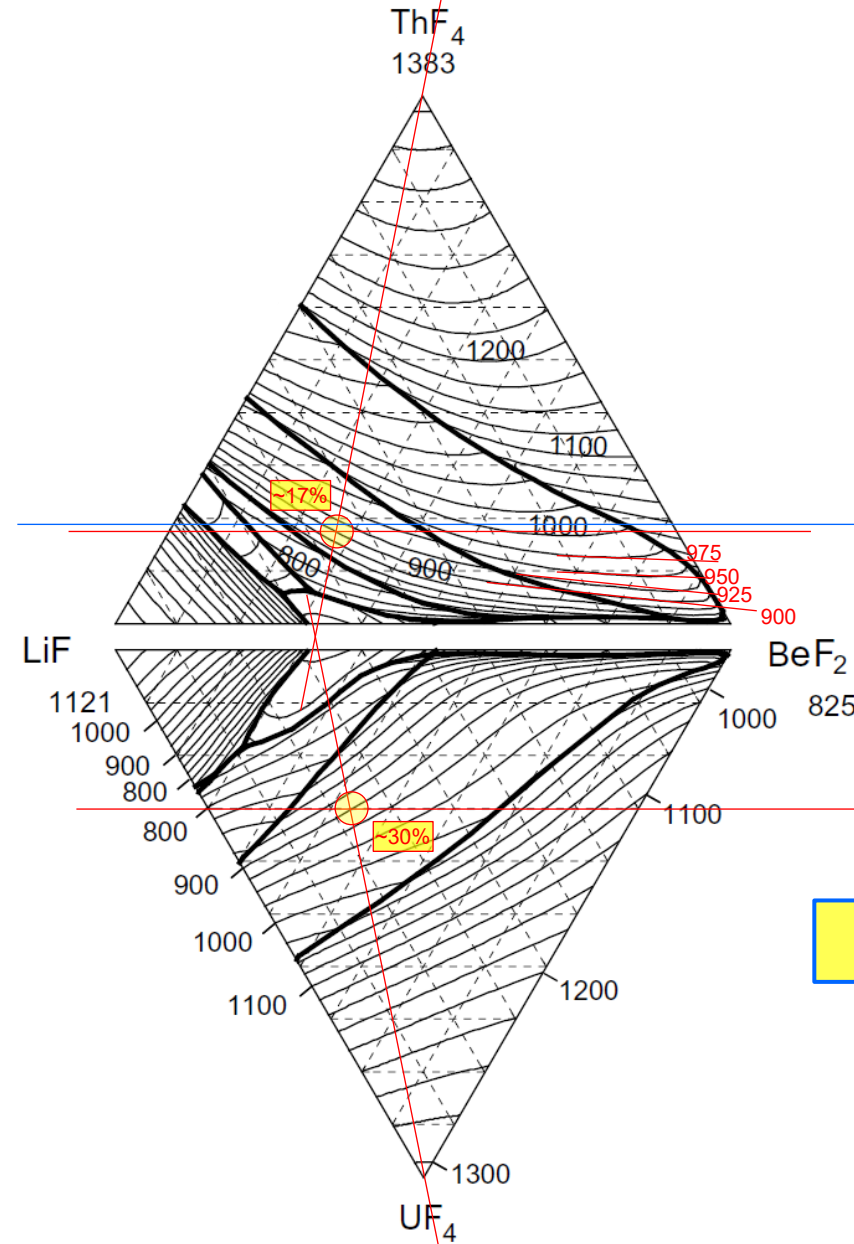


Fig. 5. Calculated liquidus surface of $\text{LiF}-\text{BeF}_2-\text{ThF}_4$, $\text{LiF}-\text{BeF}_2-\text{UF}_4$, $\text{LiF}-\text{ThF}_4-\text{UF}_4$ and $\text{BeF}_2-\text{ThF}_4-\text{UF}_4$, combined to a quaternary system. Melting temperatures are labelled in K and isotherms with an interval of 25 K are shown.

How about for ThF₄?

van der Meer, *J. Nucl. Mat.* (2006):

- Used FactSage 5.3 solver with thermodynamic data from JRC-ITU (Karlsruhe/EU)
- Gets **17% ThF₄** solubility in 2(LiF)-BeF₂ at 900 K



GOOD agreement between van der Meer's FactSage/ITU vs. ORNL experimental data

No simulations by Patrick because FLiBe-ThF₄ is not prepared in MSD-TC yet!

Fig. 5. Calculated liquidus surface of LiF-BeF₂-ThF₄, LiF-BeF₂-UF₄, LiF-ThF₄-UF₄ and BeF₂-ThF₄-UF₄, combined to a quaternary system. Melting temperatures are labelled in K and isotherms with an interval of 25 K are shown.

Liquidus isoclines for LiF-BeF₂-UF₄ system

Generated using ORNL Thermochemica
software and MSD-TC V4.1 data

E = EUTECTIC
 P = PERITECTIC
 UF_4 = PRIMARY PHASE FIELD

Fixed LiF:BeF₂ ratios
— 2(LiF)-BeF₂ (2:1)
— LiF-Be₂ (1:1)

Overlay: Koger 72 (ORNL-4188)
Underlay: Patrick's plot in **Celsius**

Solver parameters

Data: MSD-TC V4.1 (Fluorides, No_Func variant)

Solver: Thermochemica

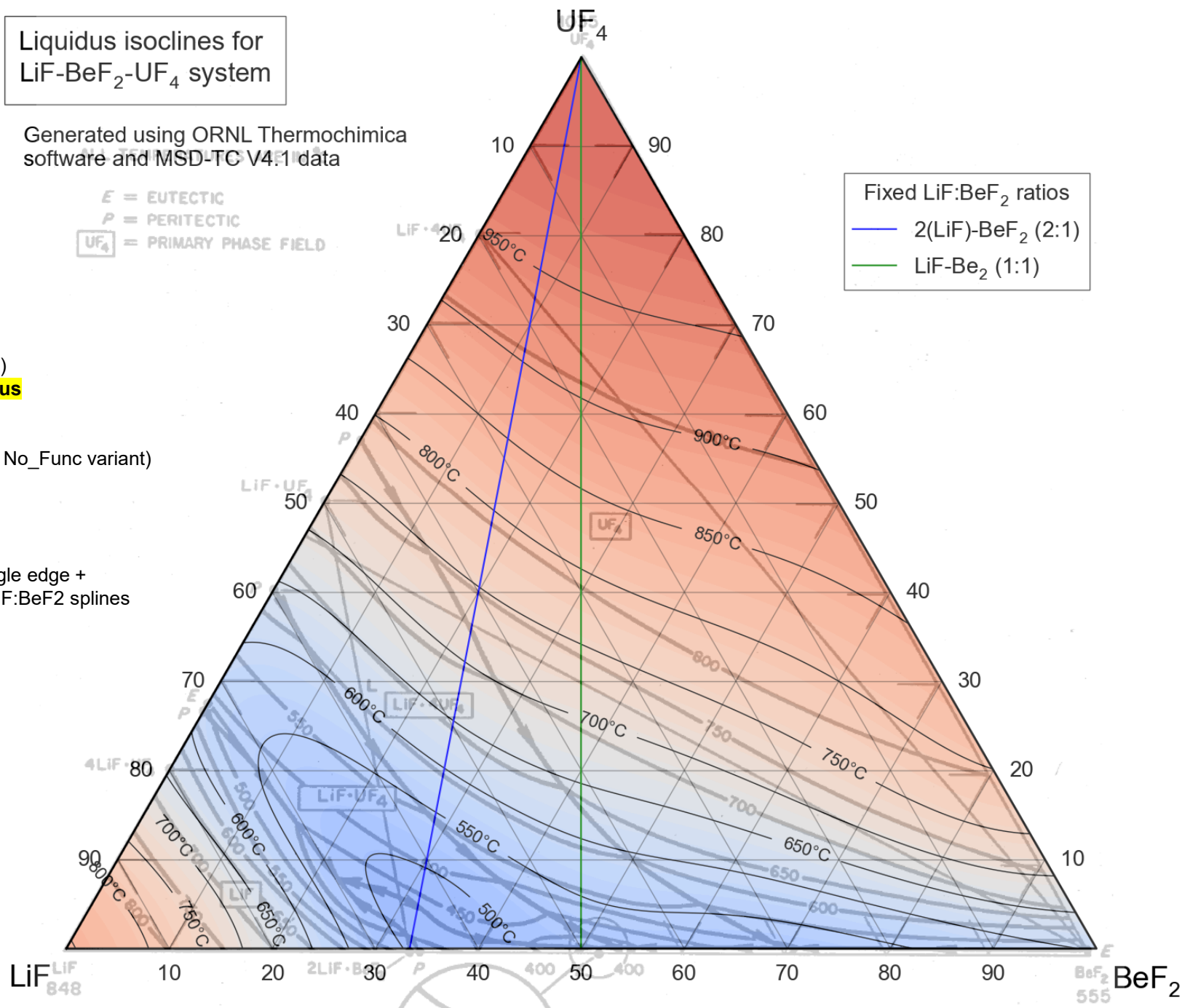
Pressure: 1 atm

Units: mol%

Temp range: 700 – 1400 K

Resolution: 120 points per triangle edge +
4x samples along 2:1 and 1:1 LiF:BeF₂ splines

PJP 2026-04-12



Liquidus isoclines for LiF-BeF₂-UF₄ system

Data: MSD-TC V4.1 (Fluorides-No_Func)
 Solver: Thermochemica (ORNL)
 Pressure: 1 atm | Units: mol%
 Resolution: 8,000 samples
 (grid of 120 points/side
 + 480 points each on 2:1
 and 1:1 LiF-BeF₂ spines)

Overlay: Koger 72 (ORNL-4188)
 Underlay: Patrick's plot in **Kelvin**

Solver parameters

Data: MSD-TC V4.1 (Fluorides, No_Func variant)
 Solver: Thermochemica
 Pressure: 1 atm
 Units: mol%
 Temp range: 700 – 1400 K
 Resolution: 120 points per triangle edge +
 4x samples along 2:1 and 1:1 LiF:BeF₂ splines

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